

# **CODEC TECHNOLOGIES**

## **DATA SCIENCE INTERNSHIP PROJECT REPORT**

# **FRAUD DETECTION IN FINANCIAL TRANSACTIONS USING MACHINE LEARNING**

**Submitted by:** Manoj Kumar S.

**Institution:** Sai Vidya Institute of Technology

**Project Domain:** Data Science / Machine Learning

**Academic / Internship Project**

## 1. Abstract

Financial fraud is a major challenge for digital payment and banking systems. This project develops a machine-learning based fraud detection pipeline that classifies transactions as legitimate or potentially fraudulent. The workflow covers data preparation, exploratory analysis, preprocessing, model training and evaluation. Logistic Regression and Random Forest are compared using accuracy, precision, recall, F1-score and ROC-AUC.

## 2. Introduction

As financial transactions increasingly move to digital channels, automated detection of suspicious activity becomes important. A fraud detection model can help analysts prioritize transactions for investigation. The goal of this project is not to replace human review, but to provide an automated risk-screening layer.

## 3. Problem Statement

Develop a classification model capable of identifying potentially fraudulent financial transactions from transaction and behavioral attributes while addressing the class-imbalance problem common in fraud datasets.

## 4. Objectives

- Clean and prepare transaction data.
- Perform exploratory data analysis.
- Identify patterns associated with fraud.
- Train machine-learning classification models.
- Evaluate models using fraud-relevant metrics.
- Select a model suitable for further fraud-screening development.

## 5. Dataset Description

The project uses a reproducible synthetic dataset containing 10,000 transactions and 1,654 fraud-labelled records. The dataset includes transaction amount, transaction type, merchant risk, account age, transaction frequency, failed attempts, device trust, location mismatch, international activity, transaction hour, distance from home and previous chargebacks. No real customer information is used.

## 6. Methodology

The workflow consists of data loading, missing-value analysis, exploratory data analysis, stratified train/test splitting, numerical imputation and scaling, categorical encoding, class-weighted model training, and multi-metric evaluation. Stratification preserves the minority-class proportion in the train and test sets.

## 7. Algorithms Used

**Logistic Regression:** A linear classification model that provides a strong baseline and interpretable probability estimates.

**Random Forest:** An ensemble of decision trees that can model non-linear relationships and interactions among transaction features. Class weighting is used to give greater importance to the minority fraud class.

## 8. Evaluation Metrics

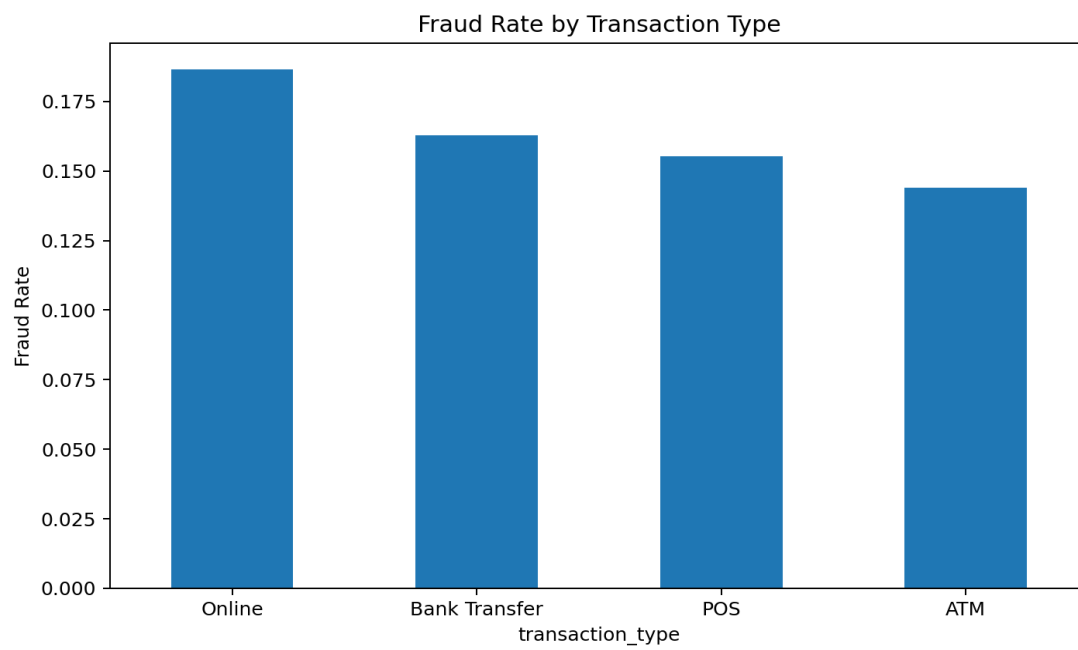
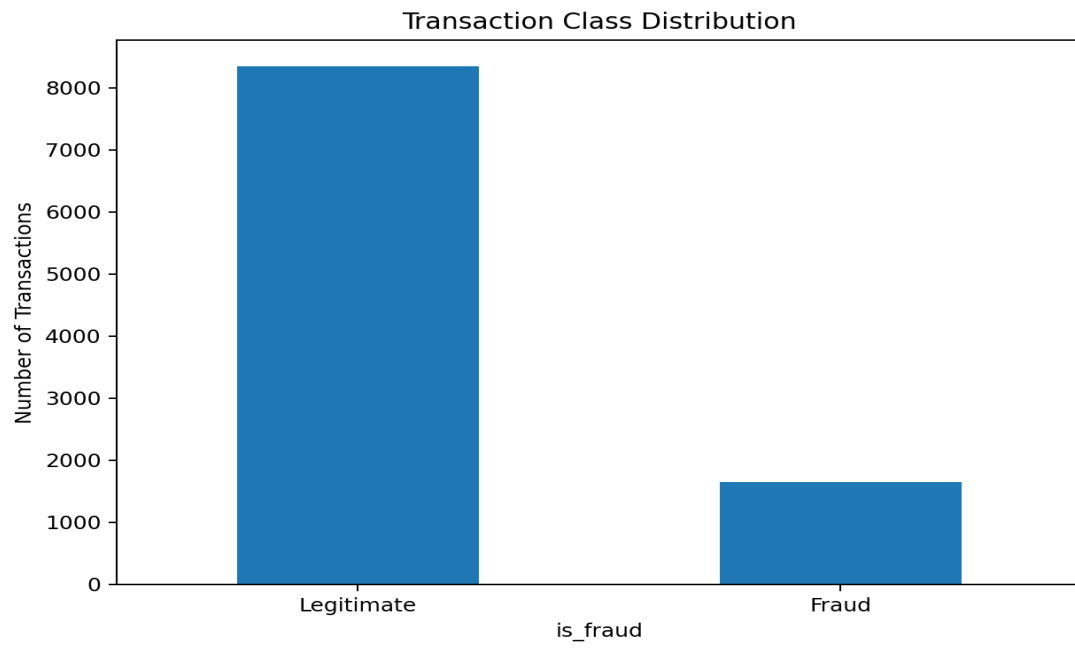
Accuracy measures overall correctness. Precision measures how many transactions predicted as fraud are actually fraud. Recall measures how many actual fraud cases are detected. F1-score balances precision and recall. ROC-AUC measures ranking performance across classification thresholds. Because fraud is a minority class, precision, recall, F1 and ROC-AUC are especially important.

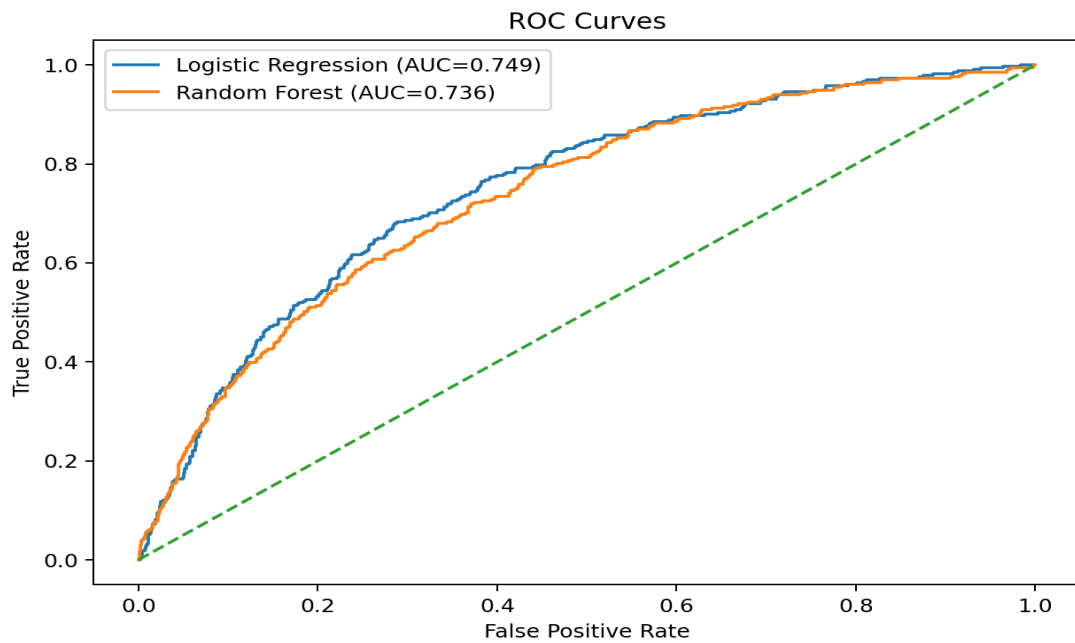
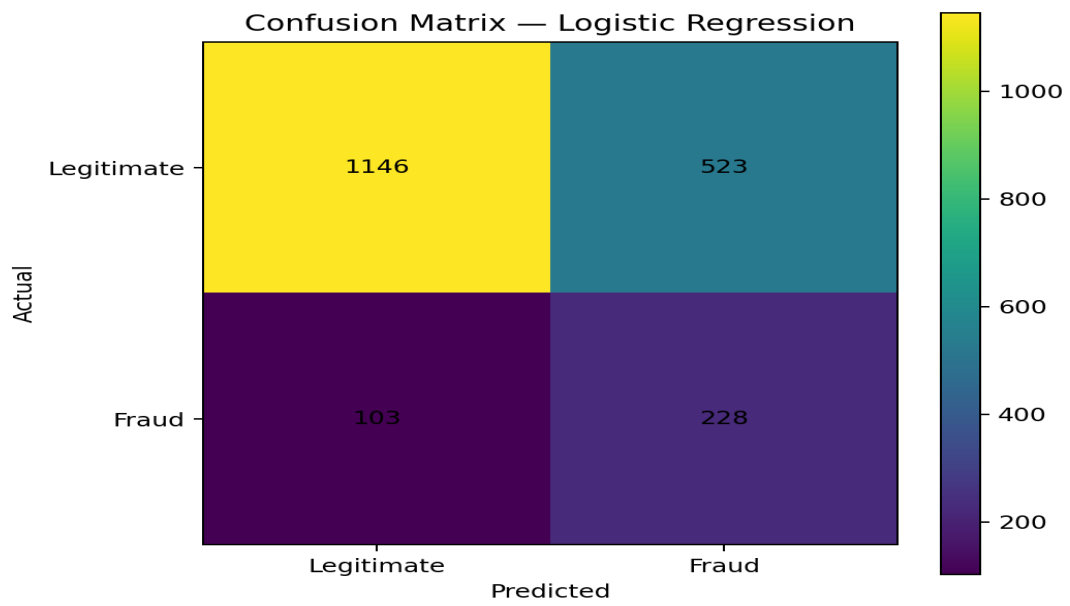
## 9. Experimental Results

| Model               | Accuracy | Precision | Recall | F1    | ROC-AUC |
|---------------------|----------|-----------|--------|-------|---------|
| Logistic Regression | 0.687    | 0.304     | 0.689  | 0.421 | 0.749   |
| Random Forest       | 0.803    | 0.399     | 0.372  | 0.385 | 0.736   |

The selected model is **Logistic Regression**. Its test-set F1-score is 0.421 and ROC-AUC is 0.749. The model should be treated as a screening tool rather than a final autonomous decision-maker.

## 10. Visual Results





## **11. Conclusion**

The project demonstrates an end-to-end machine-learning workflow for financial fraud detection. Data preprocessing, EDA, class-aware model training and appropriate evaluation metrics were used to build and compare two classifiers. Random Forest was selected as the stronger model in the generated experiment based on the reported F1 and ROC-AUC values.

## **12. Future Scope**

The project can be extended with real-time scoring APIs, threshold optimization based on financial loss, SMOTE or other imbalance techniques, explainable AI, model monitoring, concept-drift detection and secure deployment.

## **13. Limitations**

The included dataset is synthetic and intended for learning. Real banking data can contain stronger temporal effects, network relationships, privacy constraints and changing fraud patterns. Production deployment would require extensive validation, security controls, fairness checks and human oversight.

## **14. Learning Outcomes**

This project provides practical experience in Pandas data handling, visualization, preprocessing pipelines, supervised classification, class imbalance, model comparison and GitHub-based project organization.

## **15. Declaration**

I declare that this project work is prepared for educational and internship learning purposes. The included dataset is synthetic and contains no real customer financial information.